

DTSA 5003

Hypothesis Testing for Data Science

|                                                                        |                          |               |
|------------------------------------------------------------------------|--------------------------|---------------|
| FORMAT                                                                 | CATEGORY                 | CREDITS       |
| Online (DTSA)                                                          | Core                     | 1             |
| SECTION                                                                | BEST ROLE MATCH          | AVG ALIGNMENT |
| Section 1 – Statistics Pathway                                         | AI Engineer (61.0 / 100) | 50.6 / 100    |
| SPECIALIZATION Data Science Foundations: Statistical Inference Pathway |                          |               |

51

CURRENT AVG SCORE  
across 6 job roles

80

POTENTIAL SCORE  
with recommended additions

+35

MAX SCORE GAIN  
best single-role improvement

CURRENT COURSE DESCRIPTION

This course will focus on theory and implementation of hypothesis testing, especially as it relates to applications in data science. Students will learn to use hypothesis tests to make informed decisions from data. Special attention will be given to the general logic of hypothesis testing, error and error rates, power, simulation, and the correct computation and interpretation of p-values. Attention will also be given to the misuse of testing concepts, especially p-values, and the ethical implications of such misuse.

INDUSTRY ALIGNMENT BY JOB ROLE

| JOB ROLE                  | CURRENT SCORE               | PROJECTED SCORE             | POINT GAIN | PRIORITY |
|---------------------------|-----------------------------|-----------------------------|------------|----------|
| AI Engineer               | <div><div></div></div> 61.0 | <div><div></div></div> 85.3 | +24.3      | MEDIUM   |
| Machine Learning Engineer | <div><div></div></div> 59.3 | <div><div></div></div> 88.6 | +29.3      | MEDIUM   |
| ML Engineer               | <div><div></div></div> 49.2 | <div><div></div></div> 75.3 | +26.1      | MEDIUM   |
| MLOps Engineer            | <div><div></div></div> 48.1 | <div><div></div></div> 82.6 | +34.6      | HIGH     |
| Applied Scientist         | <div><div></div></div> 47.4 | <div><div></div></div> 76.1 | +28.7      | MEDIUM   |
| Data Scientist            | <div><div></div></div> 38.6 | <div><div></div></div> 73.7 | +35.1      | HIGH     |

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold ( **example** ) appear as gaps across 3 or more job roles.

**System Design**

4 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

HIGH

Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

GAP KEYWORDS BY ROLE

|                                                                      |                         |
|----------------------------------------------------------------------|-------------------------|
| <b>AI Engineer</b>                                                   | 61.0 → 85.3 (+24.3 pts) |
| models learning software model systems based problems machine        |                         |
| <b>Machine Learning Engineer</b>                                     | 59.3 → 88.6 (+29.3 pts) |
| sets learning tools exploratory design models model machine          |                         |
| <b>ML Engineer</b>                                                   | 49.2 → 75.3 (+26.1 pts) |
| media learning machine systems models model world problems           |                         |
| <b>MLOps Engineer</b>                                                | 48.1 → 82.6 (+34.6 pts) |
| model scientists systems learning machine media tools models         |                         |
| <b>Applied Scientist</b>                                             | 47.4 → 76.1 (+28.7 pts) |
| computational problems models learning systems industry design world |                         |
| <b>Data Scientist</b>                                                | 38.6 → 73.7 (+35.1 pts) |
| rules sets simply problems                                           |                         |

**Methodology:** Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.